

Concepts spread where they stick: network signals of interdisciplinary diffusion in science

Abstract

New scientific concepts differ in how broadly they spread across disciplines. We ask which early structural signals in a concept’s topic co-occurrence network anticipate later cross field breadth (the breadth of a concept’s diffusion across distinct scientific fields) and by what routes concepts move between fields. Using the full OpenAlex snapshot, we identify thousands of concepts by title matching, track their adoption episodes across scientific fields, and screen early network indicators against a popularity baseline on held out field groups never used in selection. Seven indicators survive multiple testing correction, and a learned model combining them exceeds the baseline by +0.059 in Spearman correlation for predicting rarefied cross field breadth. The retained frontier hypothesis, that concepts next enter fields related to the ones currently retaining them, is confirmed in a preregistered held out test: conditional logit coefficient $d_0 = 0.322$ [0.291, 0.355], positive in all four domain groups. This extends the principle of relatedness from economic geography to science, with the novel element that retention, not merely initial adoption, predicts where a concept goes next. A decomposition shows that 73% of the breadth gap between integrating and localised concepts is early contact diversity, while frontier advance contributes near zero. The signal is topical nonredundancy of the early neighbourhood, not temporal partner turnover.

1 Introduction

Some scientific concepts stay within their home discipline for decades, while others cross field boundaries within a few years. Optogenetics, originating in neuroscience, entered genetics, psychiatry and bioengineering; deep learning, rooted in computer science, now appears in medicine, materials science and linguistics. Understanding what distinguishes broadly diffusing concepts from locally absorbed ones matters for science policy, research evaluation and the design of interdisciplinary programmes [Cheng et al., 2023, Hofstra et al., 2019]. If early structural signals predict later cross field integration, funders and institutions could identify concepts with high diffusion potential before that diffusion is observable in publication counts alone.

The problem is hard because simple popularity metrics, including publication volume, growth rate and off home publication share, already explain much of the variance in later breadth [Rotolo et al., 2015]. Any network indicator must add information beyond these baselines. Prior work has proposed several candidate signals: community diversity of early adopters [Weng et al., 2013], collaboration density before emergence [Salatino et al., 2017], and structural variation in citation networks [Salatino et al., 2018]. These studies typically operate on curated concept sets without held out validation, mix units of analysis defined at the level of individual concepts with those at the field level, or do not control for volume based predictors. No prior study has screened a broad set of temporal knowledge network indicators against a shared baseline on held out field groups for the specific outcome of size adjusted interdisciplinary breadth.

A second open question concerns the routes by which concepts move between fields. The principle of relatedness, established for regional economic diversification, predicts that actors diversify into activities related to their existing portfolio. For scientific fields adopting new concepts, this principle has been tested for entry but not for the role of currently retaining fields, those that have adopted a concept and kept publishing on it. Whether concepts spread next to fields related to the ones currently retaining them, beyond relatedness to the home field alone, is untested.

This paper addresses both questions with a large scale empirical study. We identify 12,499 concepts from the OpenAlex bulk snapshot (476 million works), compute 53 early network indicators across seven families, and test them on held out field groups and a confirmatory onset cohort. We build a conditional logit model, a discrete choice model that estimates the probability of entry into each candidate field within a stratified choice set, to test whether retained field relatedness predicts the next field a concept enters. We decompose the breadth gap between integrating and localised concepts into early contact, frontier advance and retention channels.

Our contributions are:

1. We screen 53 early co-occurrence network indicators against a five feature popularity baseline and confirm seven on held out field groups for predicting rarefied cross field breadth, with an ElasticNet combining them exceeding the baseline by +0.059 [+0.046, +0.073] in Spearman correlation.
2. We confirm, in a preregistered test on 3,162 held out concepts, that concepts spread next to fields related to the ones currently retaining them (conditional logit $d_0 = 0.322$ [0.291, 0.355]), extending the principle of relatedness to concept diffusion in science.
3. We decompose the breadth gap between widely and narrowly spreading concepts: 73% is early contact diversity, near 0% is frontier advance, and 25% is differential retention.
4. We show that early neighbourhood consolidation, measured by ideational consistency, predicts volume growth but narrower cross field reach, a reversal that holds in all five domain groups.
5. We replicate the openness signal on a fresh 2015–2017 cohort (partial Spearman +0.091 [+0.013, +0.171]) and on vocabulary free newborn phrases.

The remainder of the paper is organised as follows. Section 2 surveys related work. Section 3 describes the data, indicators and models. Section 4 presents results. Section 5 discusses the findings, and Section 6 states limitations.

2 Related Work

Concept diffusion in science. Cheng et al. [2023] tracked roughly 2,000 new ideas across Web of Science fields and found that early social reach, the number of disconnected researcher groups adopting an idea, was the strongest predictor of whether an idea became core. Their ideational consistency measure, the cosine similarity of vectors recording which topics a concept’s neighbours use across consecutive years, predicted next year volume. Rotolo et al. [2015] defined emerging technologies through five attributes including novelty and fast growth but did not test cross field spread. Salatino et al. [2017] showed that collaboration density rises before topic emergence. These studies provide the starting point for our work but do not screen a broad indicator set against a shared baseline with held out validation for size adjusted cross field breadth.

Topic co-occurrence and citation networks. Callon et al. [1991] introduced co-word analysis as a tool for mapping the structure of research fields. Chavalarias and Cointet [2013] found that dense term clusters survive longer, and that density rises during emergence and falls before decline. Salatino et al. [2018] proposed the AUGUR system for detecting topic emergence from co-occurrence backbone dynamics. Our work differs in that we test co-occurrence indicators as predictors of a specific outcome, cross field breadth, rather than using them descriptively.

Novelty, recombination and impact. Uzzi et al. [2013] found that high impact papers combine conventional and atypical journal pairings. Hofstra et al. [2019] documented the diversity–innovation paradox: researchers from underrepresented groups produce more novel work but receive fewer citations. Tria et al. [2013] modelled the dynamics of correlated novelties, showing that novelty begets novelty through the adjacent possible. Burt [2004] showed that structural holes, brokerage positions spanning disconnected groups, generate good ideas. These studies operate at the paper or individual level; ours operates at the concept level and asks about cross field breadth rather than citation impact.

The principle of relatedness. Hidalgo et al. [2007] showed that a country’s position in the product space predicts which products it diversifies into. Neffke et al. [2011] extended this to regional industry diversification. Our contribution is to test whether relatedness to the set of fields currently retaining a concept predicts entry, beyond relatedness to the home field and conventional density measures.

Community structure and virality. Weng et al. [2013] showed that early community diversity predicts virality of online content (AUC = 0.83). Ugander et al. [2012] found that structural diversity drives social contagion on Facebook. Centola [2010] demonstrated experimentally that complex contagions require reinforcement from multiple communities. Palla et al. [2007] tracked the evolution of overlapping communities in large networks. Our indicator screen tests the scholarly analogue of these mechanisms: whether a concept’s early co-occurrence partners spanning multiple communities predicts later cross field breadth.

3 Data and Methods

3.1 Concept identification and panel construction

We use the OpenAlex bulk snapshot (476,196,327 works; 129.4 million base works 1995–2022). Concepts are identified by Aho–Corasick title matching of 56,643 legacy OpenAlex concepts (levels 2–5) plus Wikidata aliases, with stemmed verification. A per concept precision gate drops concepts with precision below 0.80 (benchmark precision 0.947, recall 0.659, assessed on a benchmark of 390 pairs with 60 hand checked pairs at 90% agreement).

A concept’s onset year t_0 is the first year it reaches 20 grounded publications. Home fields are the fields holding at least 40% of the first 30 grounded papers. The frame comprises 12,499 concepts and 27,393 adoption episodes (one per concept–field pair).

The panel is split by home field family into a development set (DEV: Computer Science, Engineering, Biochemistry/Genetics/Medicine; 4,771 concepts), four held out field groups (Physical Sciences 742, Life and Environment 1,113, Social Sciences 1,352, Mathematics and Decision Sciences 165), and a 2010–2014 onset cohort (4,356 concepts, all fields). The specification was hash sealed on DEV before held out scoring.

3.2 Field backbone

The interfield backbone is a weighted graph over 26 fields, with edge weights given by positive pointwise mutual information (PMI) of topics appearing jointly in the same work within a preonset window (1998–2002). This backbone is used throughout for relatedness density, gateway centrality and community detection. Its time varying versions (computed in five year slices) correlate at Spearman 0.92 with the frozen version.

3.3 Outcome: rarefied cross field breadth

The primary outcome is rarefied cross field breadth O_{2r} ($m = 50$): the expected number of distinct venue fields among a fixed size random draw of 50 papers from a concept’s publications in years $t_0 + 6$ to $t_0 + 8$, computed by exact hypergeometric rarefaction. Rarefaction separates size adjusted breadth from sheer volume. Secondary outcomes include sustained uptake, transience, and field- and year normalised citation growth.

3.4 Baseline

The five feature popularity baseline is: log publication volume at $t_0 + 2$, publication growth rate, log off home volume, off home share, and number of home fields. This baseline alone reaches Spearman correlations of 0.65–0.86 with O_{2r} across domain groups, setting a high bar for incremental network indicators.

3.5 Indicator families

We compute 53 indicators across seven families in the t_0 to $t_0 + 2$ window:

- **Popularity (E):** publication growth, author growth, author count.
- **Disciplinary composition (F):** Shannon entropy of field distribution, home field relatedness, Rao–Stirling diversity.
- **Landing position (G):** gateway centrality of home and early off home fields.
- **Retained frontier (FR):** relatedness density to fields with revealed comparative advantage (RCA) > 1 , volume weighted density, contact reach (count of off home fields entered), retention ratio (share of contacted fields that retain the concept).
- **Co-occurrence ego network (A):** 27 indicators from the concept’s ego network on the co-occurrence backbone, including neighbourhood novelty (share of new partners), edge persistence (share of retained partners), number of Leiden communities among neighbours, ego density, participation coefficient, brokerage.
- **Author overlap ties (S):** shared authorship between concepts.
- **External recognition:** whether the concept appears in domain taxonomies or curated lists.

3.6 Indicator selection and validation

Indicators are screened on DEV by partial Spearman correlation with O_{2r} given the popularity baseline, using leaving one home group out cross validation with 2,000 bootstrap resamples of concepts. The top 10 frozen indicators per outcome are evaluated on held out groups with DerSimonian–Laird random effects pooling across four domain groups and Holm–Bonferroni correction for multiplicity (a stepdown procedure controlling the familywise error rate).

3.7 Retained frontier model

For the field entry analysis, we construct risk sets indexed by concept, target field and year: at each year after onset, every off home field that the concept has not yet entered is at risk. Entry is defined as the concept reaching 5 grounded publications in the field. The conditional logit (Breslow partial likelihood) is stratified by each combination of concept and year, with standardised covariates added in four nested rungs: the baseline includes relatedness to home, log field size, entered field density based on revealed comparative advantage and own field gateway centrality; the second rung adds density of fields with revealed comparative advantage above unity; the third adds volume weighted density; and the full model adds retained field relatedness d_0 , defined as the mean backbone relatedness between the target field and the set of off home fields currently retaining the concept. The specification is frozen on DEV and scored once on the held out frame.

3.8 Breadth decomposition

We decompose the gap in retained off home breadth at $t_0 + 8$ between the top and bottom O_{2r} terciles into three multiplicative factors: early contact E_2 (fields entered by $t_0 + 2$), frontier advance M (ratio of fields entered by $t_0 + 8$ to those entered by $t_0 + 2$), and retention ratio ρ (share of entered fields retained at $t_0 + 8$). The decomposition is volume stratified.

4 Results

4.1 Which indicators predict cross field breadth?

Seven of the 53 screened indicators survive held out validation with Holm–Bonferroni correction for predicting rarefied cross field breadth (O_{2r} , $m = 50$), all positive in six of six domain groups and cohort bodies (Table 1).

The first two indicators, cumulative density and volume weighted density, use the cumulative field history up to $t_0 + 2$ and partly reflect a preonset footprint. After removing the preonset component, the postonset volume weighted density attenuates to about half its raw value (+0.176 vs +0.307) and is nearly rank identical to the popularity baseline (Spearman between the two > 0.97). The remaining indicators, contact reach, community count, novelty, retention ratio and ego density, measure genuinely early structural properties.

A learned ElasticNet combining the confirmed indicators exceeds the popularity baseline by +0.059 [+0.046, +0.073] in Spearman correlation on the pooled held out set. Held out increments by domain group are: Physical Sciences +0.050, Life and Environment +0.059, Social Sciences +0.062, Mathematics and Decision Sciences +0.037.

For sustained uptake, only the number of early authors is confirmed. For citation growth, home field relatedness (negative) and author growth are confirmed but the learned model reaches only Spearman 0.188 against the popularity baseline’s 0.015. External recognition shows no association with any network indicator beyond onset year.

Table 1: Confirmed indicators for rarefied cross field breadth (O_{2r} , $m = 50$). Pooled partial Spearman correlations from DerSimonian–Laird random effects pooling across four held out domain groups. All seven survive Holm–Bonferroni correction.

Indicator	Family	Pooled ρ	95% CI	I^2
Cumulative density of entered fields	FR	+0.375	[+0.279, +0.462]	0.74
Volume weighted density	FR	+0.307	[+0.256, +0.356]	0.10
Contact reach (off home fields entered)	FR	+0.211	[+0.161, +0.261]	0.00
Communities among co-occurrence neighbours	A	+0.167	[+0.063, +0.267]	0.78
Neighbourhood novelty	A	+0.151	[+0.044, +0.255]	0.75
Retention ratio of contacted fields	FR	−0.114	[−0.160, −0.067]	0.00
Ego network density	A	−0.102	[−0.151, −0.053]	0.00

4.2 Openness composite and confirmation

The six ego network components of the confirmed indicators compose into an openness index (OPEN), defined as the mean of six z -scored components with constants frozen on the full frame of all 12,499 concepts.

On a fresh 2015–2017 onset cohort of 1,443 concepts never used in any selection step, OPEN measured within the home field (OPEN_{home}) shows a partial Spearman correlation with O_{2r} of +0.091 [+0.013, +0.171] at the second control rung (controlling for the popularity baseline, onset year and contact reach), and +0.080 [+0.001, +0.162] at the third rung (adding concept type and preonset footprint). The confidence interval includes zero at higher rungs, and the DerSimonian–Laird pool across groups is +0.083 [−0.007, +0.173]. The signal carries no forecasting gain over the popularity baseline (+0.002 Spearman). The preregistered verdict is confirmed but marginal.

An evidence synthesis across six nonselection bodies (four held out domain groups plus the 2010–2014 and 2015–2017 cohorts) gives a DerSimonian–Laird pooled OPEN_{home} partial Spearman of +0.069 [+0.038, +0.100] with $I^2 = 0$, all six bodies positive. The selection body (DEV) estimate of +0.109 shows shrinkage of 1.58× relative to the nonselection pool.

Neighbourhood novelty carries the openness signal. The pooled partial Spearman of neighbourhood novelty and churn across five nonselection bodies is +0.105 [+0.069, +0.140] with $I^2 = 0$, while edge persistence is null on the fresh cohort. Figure 1 summarises the evidence across all evaluation bodies.

4.3 Confound analysis: topical dispersion, not temporal churn

The openness signal could be an artefact of small yearly samples. We test this with four null models:

- **Fixed- n rarefaction null** (drawing $n = 10$ partners per year): neighbourhood novelty and churn retains 68% of the raw effect (pooled +0.078 vs +0.116).
- **Temporal permutation null** (randomly reordering the year labels within each concept’s own time series, 200 permutations): the temporal excess is +0.008 with split half reliability below 0.05, indicating that the temporal ordering of partner arrivals contributes almost nothing.

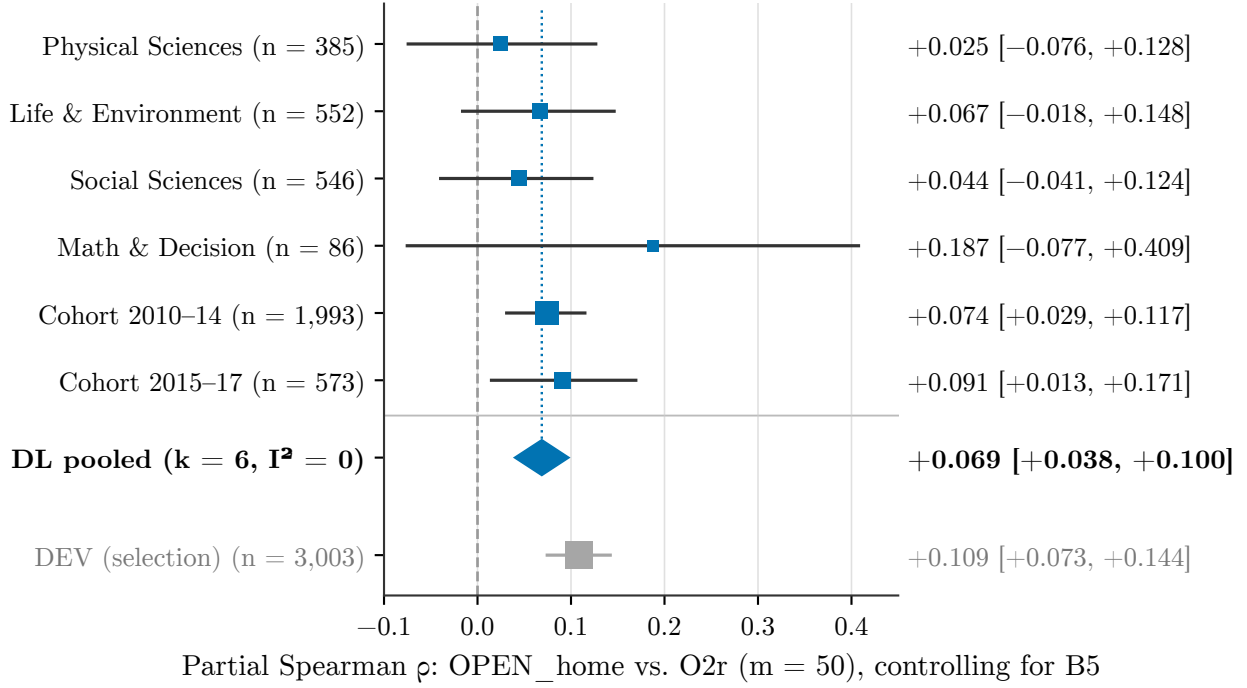


Figure 1: Partial Spearman correlation between home field openness (OPEN_home) and rarefied cross field breadth (O_{2r} , $m = 50$), controlling for the popularity baseline (second control rung), per evaluation body. Blue squares are the six nonselection bodies: four held out domain groups and the 2010–14 and 2015–17 onset cohorts. Square area is proportional to inverse variance (Fisher- z) weight, and horizontal lines are 95% bootstrap CIs (2,000 bootstrap resamples of concepts); n is the number of concepts analysed. The blue diamond spans the DerSimonian–Laird random effects pool over these six bodies, +0.069 [+0.038, +0.100] with $I^2 = 0$, and the dotted blue line marks the pooled estimate. The grey row is the DEV selection body (+0.109 [+0.073, +0.144], $n = 3,003$), on which the index was chosen. It is not part of the pool and is $1.58\times$ the pooled estimate, consistent with winner’s curse shrinkage. The dashed grey line marks zero. All six nonselection estimates are positive, but only the two cohort intervals exclude zero. Because five of the six bodies had been unsealed earlier, the pool is descriptive rather than confirmatory.

- **Configuration model null** (rewiring the ego network while preserving each node’s degree, following the Molloy–Reed configuration model): ego density normalised against the null (z -scored) still predicts breadth (partial Spearman -0.091).
- **Split half reliability**: neighbourhood novelty and churn raw = 0.48, $\text{OPEN}_{\text{home}}$ = 0.49, rising to 0.58 after configuration model cleaning.

Raw edge persistence is 66% explained by its own temporal permutation null mean (computed by reordering year labels within each concept’s time series), and that null mean predicts the outcome at least as strongly as raw persistence itself. The signal is a static topical dispersion property of the home topic mix, not temporal partner turnover.

4.4 Reconciling consistency and breadth

Cheng et al. [2023] reported that ideational consistency (cosine of a concept’s topic usage vector across consecutive years) predicts next year volume with a 53% increase per standard deviation. We reproduce this almost exactly: our twin negative binomial model gives +53.5% per SD. Adding log current volume reduces this to +1.3% [+0.5%, +2.1%]: the raw association is size dominated.

Net of size, consistency’s partial Spearman with rarefied cross field breadth is -0.069 [-0.093 , -0.047], negative in all five domain groups. On the 2015–2017 cohort, the reversal replicates at -0.111 [-0.197 , -0.030]. Panel analysis tracking each concept over time shows that years of high consistency are followed by slightly more off home entries ($b = +0.025$ [+0.004, +0.048]), so the reach penalty is a trait observed across concepts, not a mechanism operating within a single concept over time.

Consistency and breadth are not opposed within a single concept’s trajectory, but concepts whose neighbourhood is consistently structured tend to grow in volume rather than spreading across fields. This complements Cheng et al. [2023] by separating the volume and breadth outcomes.

4.5 How do concepts diffuse across fields?

4.5.1 Retained frontier entry

The conditional logit model tests whether a concept’s next field of entry is predicted by the target field’s relatedness to the off home fields that currently retain the concept, beyond relatedness to the home field, field size, entered field density and own gateway centrality.

On the development frame (274 concepts, 961 informative strata), the model reproduces a prior result: likelihood ratio 68.6, $d_0 = 0.281$.

On the independent held out frame (3,162 concepts, 6,978 entry events, hash frozen specification), the retained frontier coefficient is $d_0 = 0.322$ [0.291, 0.355] with standard errors clustered by concept (likelihood ratio 325.8 for the retained frontier term over the volume density rung, $p < 10^{-72}$). The result is positive in all four held out domain groups: Physical Sciences 0.148, Life and Environment 0.401, Social Sciences 0.297, Mathematics and Decision Sciences 0.065 (underpowered). The DerSimonian–Laird pool over four groups gives 0.243 [0.118, 0.368] with $I^2 = 0.92$, reflecting genuine heterogeneity across domains. Three permutation tests confirm robustness: shuffling which fields retain the concept ($p = 0.001$), rewiring the field backbone ($p = 0.004$), and permuting concept labels across nodes ($p = 0.003$).

The predeclared volume matched contrast, comparing entry rates of retained versus fields that were entered but not retained in the same volume cell, is null: -0.028 [-0.105 , +0.046]. The frozen verdict is therefore partial: persistence is confounded with volume. Under a minimum conditional

probability proximity [Hidalgo et al., 2007], the retained frontier coefficient reverses to -0.021 ($p = 0.012$), indicating backbone dependence. Table 2 reports all estimates by domain group.

Table 2: Retained frontier coefficient d_0 from the conditional logit model of next field entry across domain groups and the 2010–2014 cohort, with 95% confidence intervals clustered by concept.

Model	d_0 coefficient	95% CI (concept)	Likelihood ratio
Development set	0.228	[0.164, 0.291]	34.5
Held out pooled	0.322	[0.291, 0.355]	325.8
Physical Sciences	0.148	[0.078, 0.219]	–
Life & Environment	0.401	[0.342, 0.460]	–
Social Sciences	0.297	[0.246, 0.348]	–
Math & Decision Sci.	0.065	[−0.109, 0.239]	–
2010–2014 cohort	0.321	[0.292, 0.347]	–

Figure 2 shows the estimates by domain group, and Figure 3 displays the nested model ladder. The AUC within each stratum increases from 0.847 before adding the retained frontier term to 0.852 after. The increment is small because field size alone achieves AUC 0.76, density only 0.59, and the ceiling is constrained by having only 26 fields.

4.5.2 Gateway centrality does not predict retention

A separate preregistered test asks whether the adopting field’s eigenvector centrality on the backbone predicts retention. On 27,393 held out episodes, the gateway centrality increment is effectively zero: Δ -AUC -0.00001 $[-0.0006, +0.0003]$. The minimum detectable effect is 0.004. Gateway alone has AUC 0.605 on DEV versus 0.506 on held out: the DEV signal is domain specific. Verdict: disconfirmed.

4.5.3 Breadth decomposition

We decompose the gap in retained off home breadth between the top and bottom O_{2r} terciles into $\log E_2$ (early contact), $\log M$ (frontier advance) and $\log \rho$ (retention). Volume stratified, on the held out pool excluding Medicine (to avoid a domain that dominates the localised class):

- Early contact share: 73.3%
- Frontier advance share: 1.3%
- Retention share: 25.4%
- Difference (exploration minus retention): 0.492 [0.403, 0.575]

The DerSimonian–Laird pool of the exploration minus retention difference across three held out groups is 0.504 [0.329, 0.679] with $I^2 = 0.76$.

Breadth differences between concepts are set by how many fields a concept contacts early, not by how many more it enters later. Frontier advance, the ratio of fields at $t_0 + 8$ to those at $t_0 + 2$, contributes near zero because both broad and narrow concepts gain new fields at similar rates relative to their base (Figure 4).

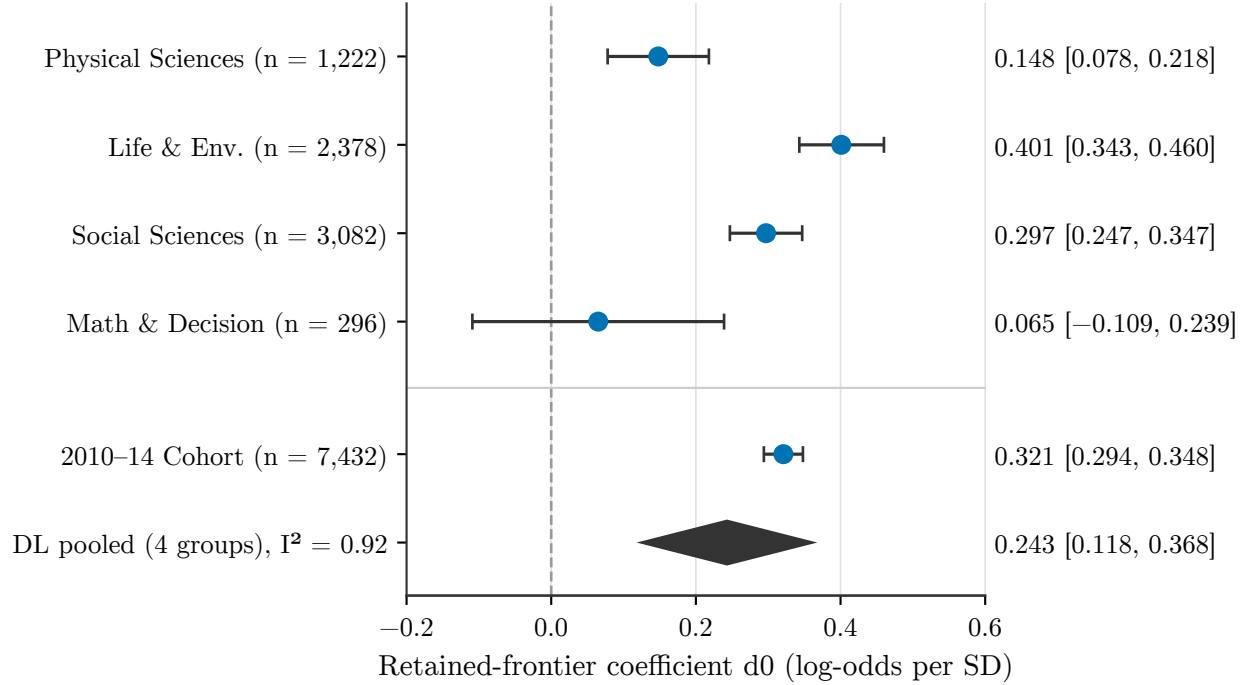


Figure 2: Retained frontier coefficient d_0 from the conditional logit model of next field entry, estimated on held out data. d_0 measures how much a target field’s relatedness to the fields currently retaining a concept predicts entry into it. It is net of relatedness to the home field, field size, entered field density and gateway centrality, and is expressed in log odds per standard deviation. Blue circles are the estimates for the four held out domain groups (top) and for the 2010–14 held out cohort. Horizontal bars are 95% Wald confidence intervals from standard errors clustered by concept, and n is the number of entry events. The dark diamond spans the DerSimonian–Laird random effects pooled estimate over the four groups, 0.243 [0.118, 0.368], with $I^2 = 0.92$. The dashed vertical line marks $d_0 = 0$, and the right hand column prints each estimate with its interval. The point estimate is positive in all four groups and in the cohort. The interval excludes zero for Physical Sciences, Life & Env., Social Sciences and the cohort. Math & Decision ($n = 296$) is the weak, imprecise group, 0.065 [−0.109, 0.239], and together with Life & Env. (0.401) it drives the high heterogeneity. These are associations. In the same experiment, a volume matched contrast between retaining and nonretaining fields is null (−0.028 [−0.105, 0.046]; not plotted), so the preregistered frontier test is only partially supported: persistence is confounded with volume.

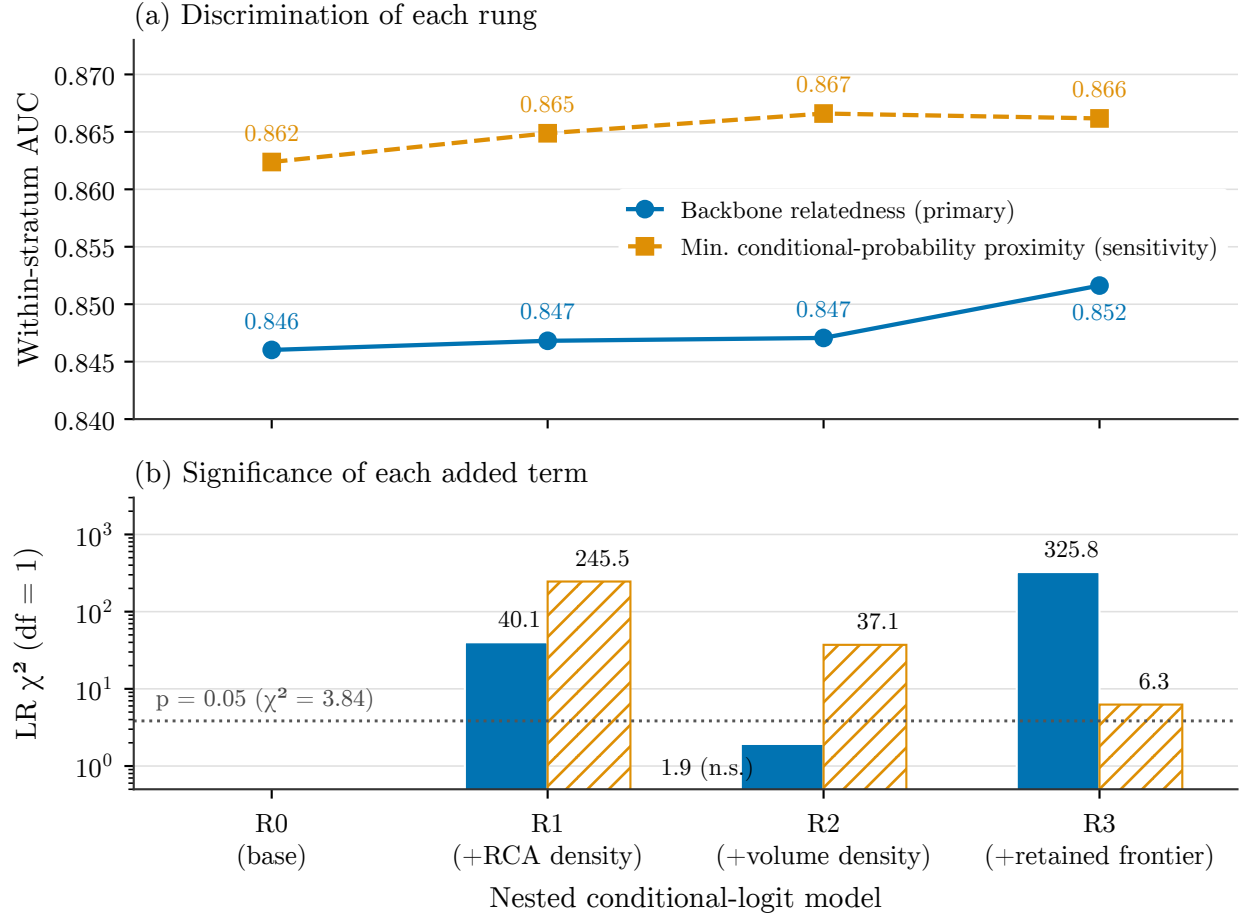


Figure 3: Nested conditional logit models of field entry on the held out frame (3,162 concepts, 6,978 entry events, 6,076 informative strata, one per combination of concept and year). The baseline rung includes home relatedness, log field size, entered field density and own field gateway centrality; successive rungs add density of fields with revealed comparative advantage above unity, then volume weighted density, then retained frontier relatedness. (a) AUC computed within each stratum for each rung. Blue circles, solid line: primary specification using backbone relatedness. Amber squares, dashed line: sensitivity rebuild using minimum conditional probability proximity. (b) Likelihood ratio χ^2 (df = 1, log scale) for the term each rung adds over the previous rung. Solid blue bars: primary. Hatched amber bars: sensitivity. The dotted line marks $p = 0.05$ ($\chi^2 = 3.84$). In the primary specification, the retained frontier term gives the largest step (LR = 325.8, $p < 10^{-72}$), but AUC rises only from 0.847 to 0.852. The volume density step is not significant (LR = 1.9). Under minimum conditional probability proximity, baseline discrimination is higher and the retained frontier term adds no AUC (0.867 to 0.866; LR = 6.3). The increment therefore depends on the proximity definition. No confidence intervals on AUC are available.

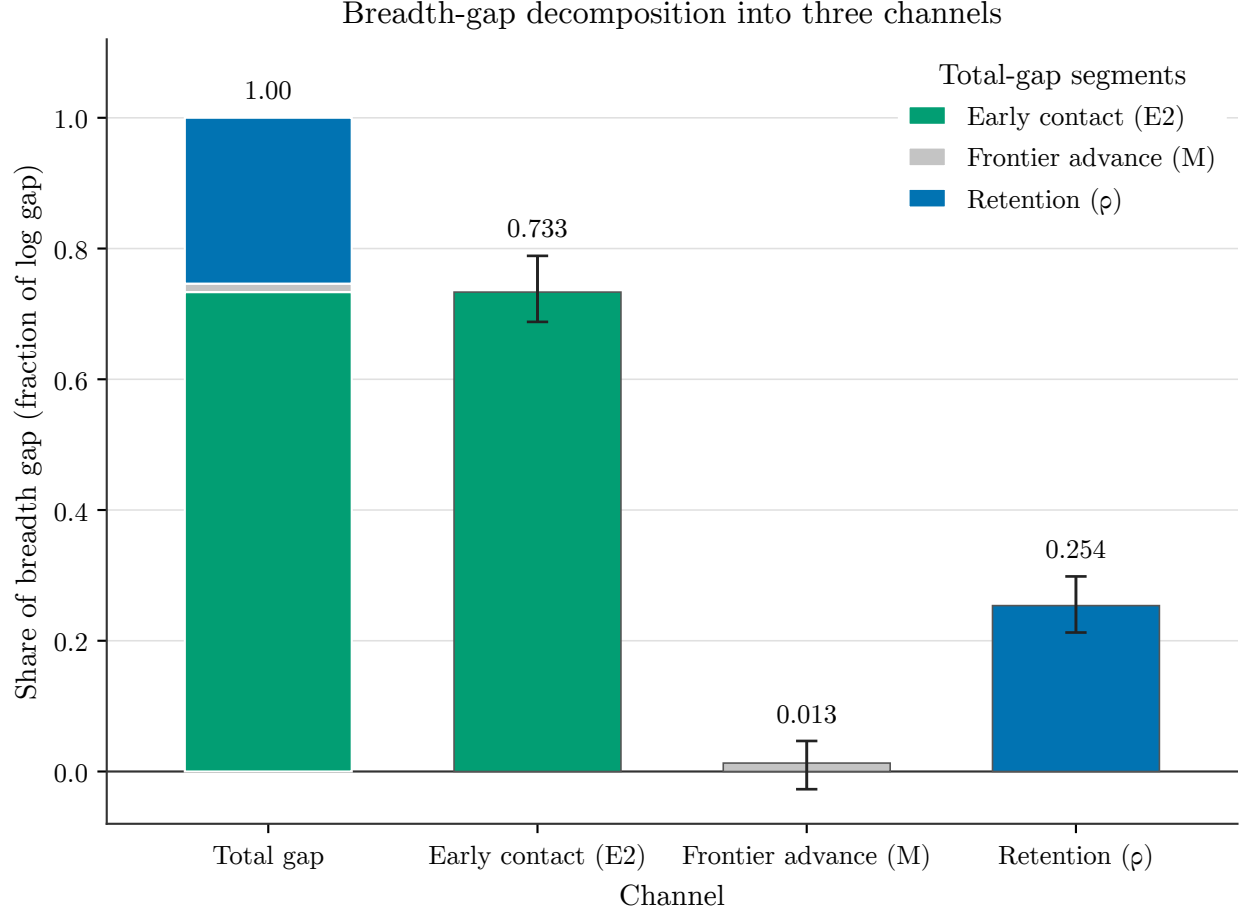


Figure 4: Decomposition of the gap in retained off home breadth (log scale) between the top and bottom O_{2r} tercile into three multiplicative channels: early contact (E_2 , green), frontier advance (M , light grey) and retention (ρ , blue). The left bar stacks the three shares into the total gap (share = 1; total log gap $D_{\text{total}} = 1.035$, 95% CI [0.957, 1.108]). The other three bars show each channel’s share of the gap with 95% percentile confidence intervals from 2,000 bootstrap resamples of concepts, with terciles and volume quintiles recomputed in each resample. The sample is the held out pool (PHYS, LIFEENV, SOC, MATHDEC), volume stratified, excluding Medicine homes ($n = 1,825$ concepts). Early contact accounts for 73% of the gap (0.733, [0.688, 0.789]), retention for 25% (0.254, [0.213, 0.298]), and frontier advance for close to zero (0.013, [-0.027, 0.047]). The shares are an accounting identity for the breadth outcome, not causal effects.

4.5.4 Diffusion trajectories: a continuum, not types

We tested whether concepts cluster into discrete trajectory types (localised emergence, rapid interdisciplinary diffusion, gradual integration) using DTW k -medoids and hidden Markov model (HMM) clustering on state sequences of field presence. No trajectory typology passes a naming rule: the adjusted Rand index between DTW $k = 4$ and HMM $S = 5$ is 0.222, and held out reclustering gives ARI = 0.44. Instead, PCA reveals a continuum: the first principal component (38.8% of variance) is a breadth axis, and the second principal component (10.7%) is a keep versus lose axis. Early openness (OPEN) correlates with the first principal component beyond the popularity baseline (held out DerSimonian–Laird partial correlation 0.120, $I^2 = 0$) but not with the second.

4.6 Replication on vocabulary free concepts

To test whether the openness signal depends on the legacy concept lexicon, we constructed a separate frame of 636 newborn title noun phrases (onsets 2003–2015) absent from the 56,643 legacy concepts. On this frame, OPEN_{home} shows a partial Spearman with rarefied breadth (O_{2r} , $m = 30$) of +0.117 [+0.020, +0.218] at the third control rung, and neighbourhood novelty and churn of +0.108 [+0.007, +0.211]. Three of four estimable domain groups are positive. The verdict is partial: the confidence interval on OPEN_{home} includes zero at the fifth control rung, and the p -value after Holm–Bonferroni correction is 0.052.

5 Discussion

Our results address both research questions posed in the Introduction.

For the first research question, seven temporal network indicators predict cross field breadth beyond a strong popularity baseline, and these indicators generalise across four held out scientific domain groups. The strongest are contact reach (the count of off home fields already entered) and neighbourhood novelty (the share of new co-occurrence partners). A learned model improves Spearman correlation by +0.059 over the baseline. The practical forecasting gain is modest: the popularity baseline already explains most of the variance, and OPEN adds only +0.002 Spearman on the fresh cohort. The signal identifies which concepts are structurally positioned for broader diffusion, rather than providing strong individual level forecasts.

The confound analysis reveals that the signal is topical nonredundancy, a static structural property of the home topic mix, rather than temporal partner turnover. A concept whose early partners span multiple communities and introduce novel topical connections tends to spread more broadly, but this tendency is set by the concept’s position in the knowledge network rather than by dynamic churn.

For the second research question, the retained frontier model confirms that concepts spread next to fields related to the ones currently retaining them. This extends the principle of relatedness from economic geography [Hidalgo et al., 2007, Neffke et al., 2011], where regions diversify into products related to their existing portfolio, to scientific concept diffusion. The novel element is retention: it is not merely which fields have adopted a concept that predicts the next field, but which of those fields continue to use it. The volume matched contrast is null, however, indicating that retention and volume are confounded and that the pure retention signal cannot be isolated at this sample size.

The breadth decomposition shows that the gap between integrating and localised concepts is set early by contact diversity (73%) rather than by continued frontier advance (1%). This echoes

Weng et al. [2013], who found that early community diversity predicts virality, and extends it from social media to scientific knowledge networks.

The reconciliation with Cheng et al. [2023] reveals an outcome dependent split: consistency predicts volume growth (size dominated) but narrower cross field reach. The two findings are compatible because volume growth and cross field breadth are partially independent outcomes.

6 Limitations

Several limitations bound the scope of these results.

First, concept identification relies on title matching against a legacy lexicon that overrepresents the natural sciences and computer science. The vocabulary free replication partially addresses this, but its sample is smaller and the verdict is partial.

Second, the field backbone uses a fixed classification of 26 fields from OpenAlex. A finer grained classification would produce more adoption episodes but also more noise in field assignment. Our results are conditioned on this specific resolution.

Third, the retained frontier claim is partial by its own preregistered rule: the volume matched contrast is null, meaning that we cannot distinguish the effect of retention from that of volume. A larger sample with more fine grained volume controls would be needed to isolate pure retention.

Fourth, the openness signal’s practical forecasting value is small (+0.002 Spearman on the fresh cohort). The indicators characterise rather than predict: they identify structural preconditions for broad diffusion but do not yield individually actionable forecasts.

Fifth, outcomes are measured at $t_0 + 6$ to $t_0 + 8$, a horizon of 6–8 years after concept onset. Longer horizons might reveal different dynamics, and our results do not speak to very long term disciplinary integration.

Sixth, the analysis is observational. Early network structure is correlated with later breadth, but we cannot establish that it causes broader diffusion. The panel analysis tracking each concept over time suggests that the association is a trait observed across concepts rather than a mechanism operating within a single concept’s trajectory.

Seventh, the held out outcomes were previously unsealed by two earlier experiments in the same project, so the indicator screen held out results are robustness checks within the same frame rather than fully independent confirmations. The fresh 2015–2017 cohort is the only body never used in any prior step.

Eighth, Mathematics and Decision Sciences is underpowered in both the indicator screen (165 concepts) and the retained frontier model ($d_0 = 0.065$, CI includes zero), so our results may not generalise to this domain.

7 Conclusion

We have shown that early structural signals in a concept’s co-occurrence network predict later cross field breadth across multiple scientific domains, that concepts spread next to fields related to the ones currently retaining them, and that early contact diversity, not continued frontier advance, accounts for most of the breadth difference between integrating and localised concepts. The openness signal is topical nonredundancy of the home neighbourhood, not temporal partner turnover. These findings connect the literatures on concept diffusion in science [Cheng et al., 2023], the principle of relatedness from economic geography [Hidalgo et al., 2007], and community structure in social contagion [Weng et al., 2013, Ugander et al., 2012], offering a network level account of why some scientific concepts become broadly integrated while others remain local.

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